

Thomas Barros

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Australian citizen

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EDUCATION

- **University of Queensland** Brisbane, QLD, Australia
Dual Bachelor of Computer Science and Bachelor of Science; GPA: 5.061/7 2022 – 2025
 - **Majors:** Physics, Machine Learning. Minor in Applied Mathematics, with further mathematics electives.
 - **Extracurricular:** UQ Computing Society (UQCS) member since 2022 and UQ Student Managed Investment Fund (UQ SMIF) challenge participant 2024 and 2025.
 - **Undergraduate Research Project:** Observing the wave-particle duality of electrons by directly observing the Friedel oscillations of copper surface state electrons using a scanning tunnelling microscope, and deriving the electron dispersion relation. Supervisor: Dr. Peter Jacobson (UQ).
- **St Philomena School** Park Ridge, QLD, Australia
High school ATAR: 98.50/99.95 2016 – 2021

SOFTWARE ENGINEERING PROJECTS

- **Solarism** – *Energy Grid Simulation and Optimisation* Available: Website, GitHub
 - Autonomous distribution system operator that optimises solar energy distribution for an energy micro-network.
 - Includes custom PVlib solar generation models with Gurobi optimisation, GridLABD energy grid simulation and React interactive dashboard front end, connected via FastAPI. Implemented with HTML, CSS, JavaScript and Python.
 - Developed by Team **solarism** with industry mentors from GridQube.
- **AI Image Generator** – *StyleGAN2 MRI Image Generator* Available: Github
 - Generates realistic MRI images of the brain. Trained on the OASIS medical image dataset.
 - Implements StyleGAN2 model architecture in Python using PyTorch, SciKit Learn, Pandas and Matplotlib libraries. Report included in the README.
- **Car Reserve** – *Car reservation app* Available: GitHub
 - Mobile app for small network of users to coordinate usage of a common car. Allows users to reserve a car for particular time periods on a calendar UI. Four monthly users as of January 2026.
 - Written with the Flutter framework in Dart with Supabase PostgreSQL backend database and user authentication.
- **Compiler** – *Teeny Tiny Compiler* Available GitHub
 - Compiler that compiles Teeny Tiny (a BASIC-like language) to C. Written in Python.
- **Sorting Algorithms** – *Web application for visualising sorting algorithms* Available: Website, GitHub
 - Visualises merge sort, quick sort, heap sort and bubble sort.
 - Written with React and Node.js in HTML, CSS and Javascript.

CERTIFICATIONS

- **IBM DevOps and Software Engineering Professional Certificate** Available: Coursera Certificate
Coursera
 - Professional certificate developing essential knowledge and skills for an entry-level DevOps role.
 - Completion of fifteen courses and projects adopting a DevOps mindset in software engineering using Agile and Scrum methodologies and Cloud Native tools and technologies.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, JavaScript, HTML/CSS, SQL, R, Dart
- **Developer Tools:** VIM, IntelliJ Idea, VS Code, Git, VirtualC, MySQL, Agile (Scrum), Jira, Drupal, Flutter